



ABSTRACT

BusinessLens is an intelligent SaaS-based analytics platform designed for businesses to provide real time analytics, data visualization, and AI-powered recommendations. The system is developed using the MERN stack and integrates OpenAI with Stripe for billing automation and data processing.

AIM & OBJECTIVE

To build an AI-powered system that automates data understanding, cleaning, and insight generation from structured business data without requiring technical expertise.



1.

Smart File Upload & Auto-Parsing

Accepts raw CSV or Excel files and automatically parses rows and columns to detect structure.

3.

Automatic Data Cleaning

Detects and corrects common data issues like nulls, formatting inconsistencies, and type mismatches.

5.

Business Insight Generation

Converts cleaned and structured data into useful business insights and summaries in plain English.

2.

Column Header Correction with LLMs

Uses OpenAI to identify and correct misnamed or missing headers based on content understanding.

4.

Semantic Column Classification

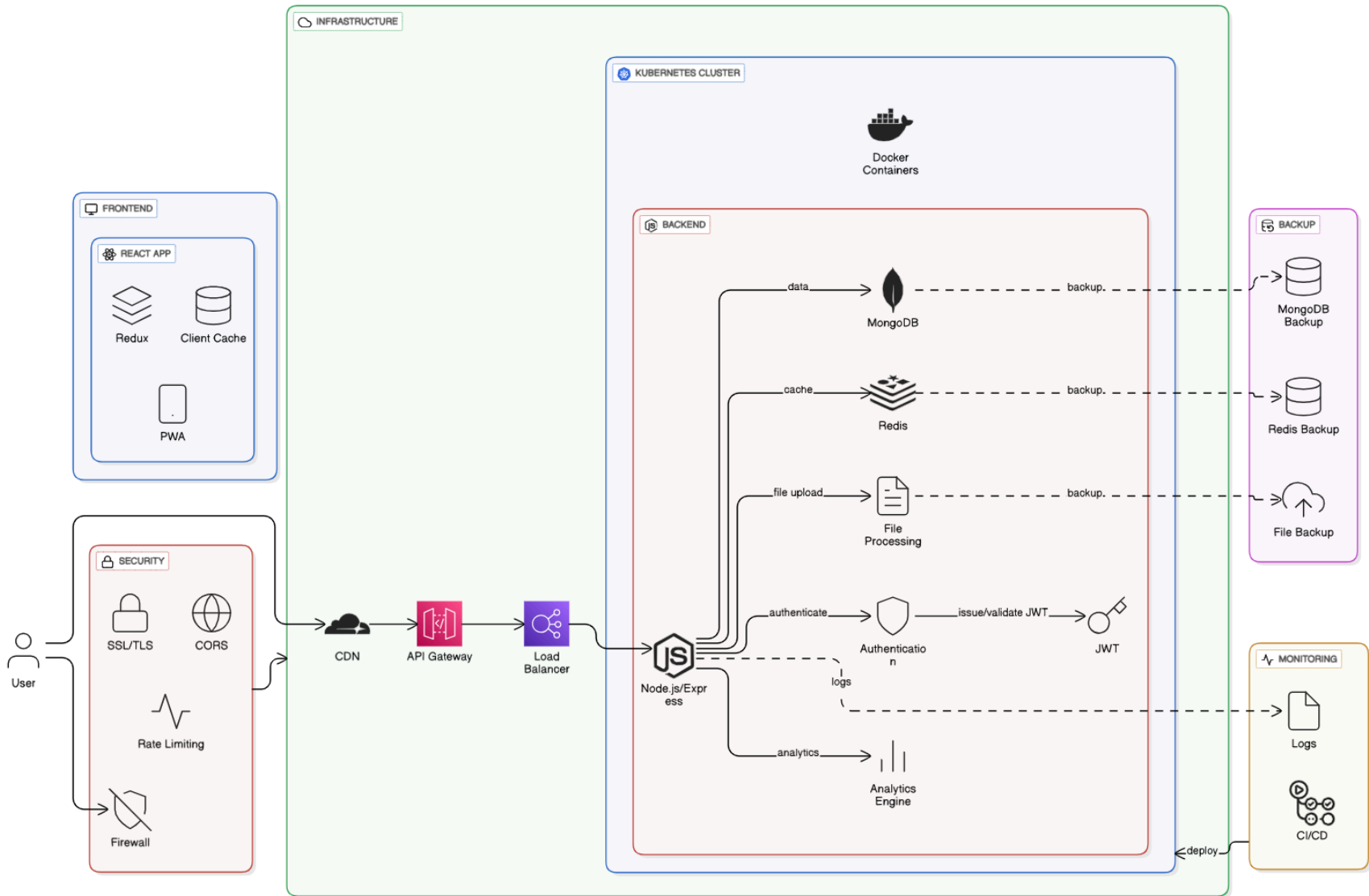
Classifies columns (e.g., metrics, dimensions, time) using contextual cues and LLM-based inference.

6.

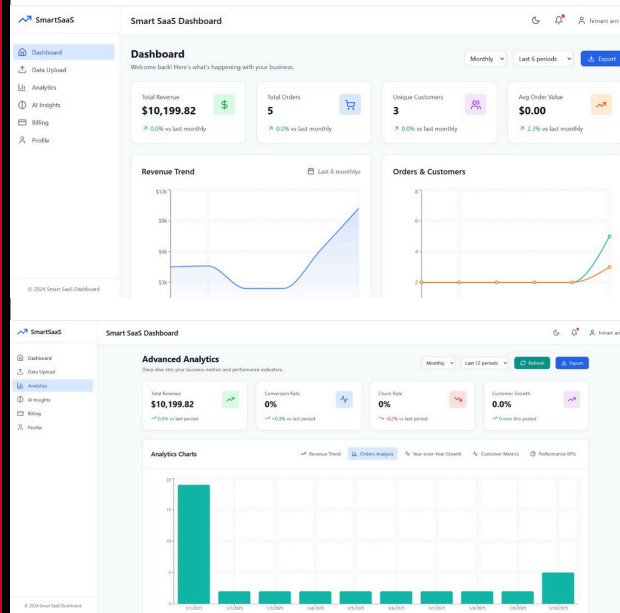
Dynamic Chart & Visualization Suggestion

Recommends appropriate chart types (e.g., bar, pie, line) based on the data's nature and distribution.

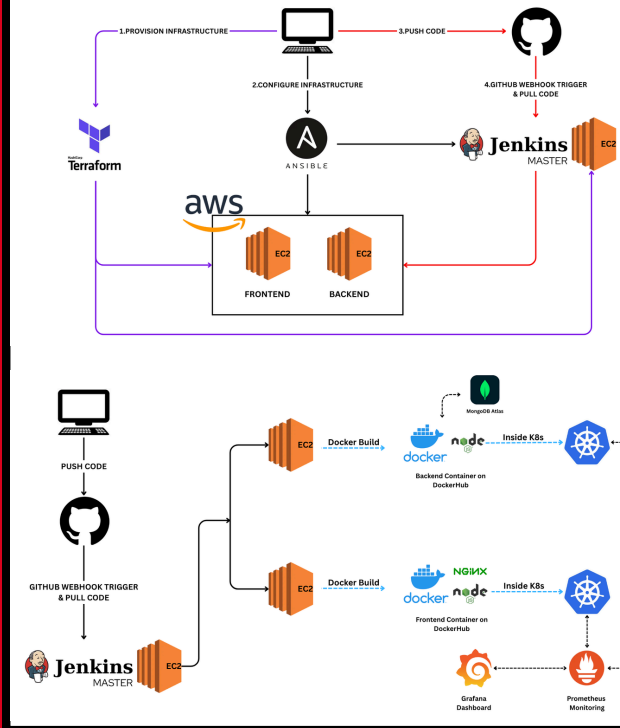
METHODOLOGY



RESULTS



DEPLOYMENT



REFERENCES

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[2] "Large Language Models (LLMs) on Tabular Data," Trans. Mach. Learn. Res., Jul. 2024. [Online]. Available: <https://arxiv.org/pdf/2402.17944.pdf>
[3] M. Parciak, B. Vandevort, F. Neven, L. Peeters, and S. Vansummenen, "Schema Matching with Large Language Models: an Experimental Study," Jul. 2024. [Online]. Available: <https://arxiv.org/abs/2407.11852>

T E C H

